

Distribution-Free Risk Control for Multi-Agent Decision Systems

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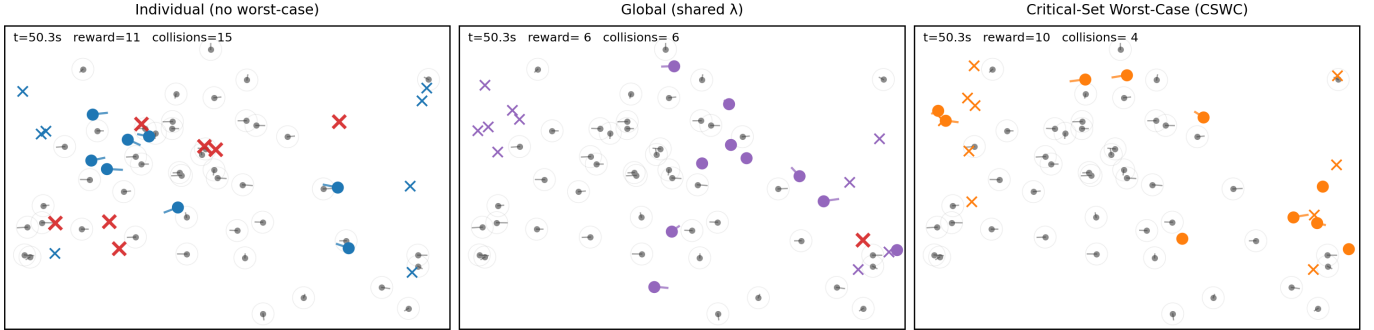


Fig. 1: Multi-Robot planner (coloured) avoiding pedestrians (gray) while trying to reach objectives (x) to collect reward. The future trajectories of pedestrians (gray) are predicted online by a machine learning algorithm (not visualised). The robots planners find an optimal spline through the scene and is penalised for being close to humans (gray outline) via an adaptive conservatism variable λ_t , which trades efficiency for safety; collision events are marked by red x. Left: independent calibration attains high throughput (11) but collides frequently (15). Middle: Global calibration across all agents suppresses collisions (6), but leads to system wide conservatism, and lower reward (6). Right (CSWC): Our proposed method updates only predicted critical channels yielding increased efficiency (reward = 10, collisions=4). Details in Section VII and videos on [project website](#)[†].

Abstract—Autonomous systems often couple learned predictions with planning, while safety is judged on realized outcomes under shift and rare failures. We study distribution-free risk control for online multi-agent decision-making with multiple safety objectives. At each time step the system selects actions from a time-varying family of decision rules indexed by a scalar conservatism parameter. Our target is the worst-channel retrospective metric: the time-average of the per-time maximum loss across all agents and objectives. We prove that independent per-agent calibration does not control this metric. We then propose a selective system-level controller that updates a single parameter using the realized worst-channel loss, but applies it only to channels that were recently maximal, leaving others unaffected. Assuming a mild delayed-safety condition, we obtain a finite-time worst-channel bound without distributional assumptions. We instantiate the theory in a drone–pedestrian navigation simulation with inertia, short-horizon prediction, and spline planning.

Index Terms—distribution-free risk control, adversarial sequences, multi-agent systems, online decision-making, safety constraints, autonomous navigation

I. INTRODUCTION

Modern autonomous decision systems increasingly couple learned prediction models with downstream optimization. A mobile robot forecasts pedestrian trajectories and then solves a trajectory planning problem; a drone predicts nearby agents and selects evasive maneuvers; a perception system predicts

uncertainty and a controller decides whether to proceed. This pipeline is powerful, but it exposes a fundamental mismatch: predictive models are trained under assumptions that rarely hold at deployment, while safety requirements are evaluated on realized outcomes under distribution shift and rare corner cases.

A standard response is to add conservatism by inflating uncertainty sets, imposing robust constraints, or choosing safety penalties that dominate worst-case prediction error. Such approaches can be effective, but they are often overly pessimistic because they treat all uncertainty as equally relevant. In many systems, only a small subset of prediction errors are *decision-relevant*: an inaccurate forecast far from the executed trajectory should not force the system to slow down. This motivates a decision-theoretic view: instead of attempting to make predictions uniformly reliable, we aim to make *decisions* reliable with respect to a specified safety metric.

When distributional assumptions are unreliable, it is natural to pursue *retrospective* (hindsight) guarantees: we ask that the deployed decisions have low realized risk on the realized deployment history, even if the data are nonstationary or adversarial. One particularly simple and attractive approach is feedback-based adaptation of a scalar *conservatism parameter* that indexes a family of decision rules from aggressive to cautious. One-dimensional feedback updates of this kind have

[†]<https://multi-agent-decision-control.github.io>

recently been shown to control empirical risk in a fully distribution-free adversarial model, even when the planner is nonconvex or heuristic [1].

Multi-agent systems rarely have a single objective. In a drone swarm navigating among pedestrians, we care simultaneously about human separation and inter-drone separation. In robot fleets, we may care about collision avoidance, workspace constraints, and throughput. A naive strategy is to tie all agents and objectives to a single shared conservatism parameter, tuned using the worst observed loss. While this yields a worst-case guarantee, it couples the entire system: a rare near-collision in one corner can force all agents into conservative behavior. Conversely, maintaining separate parameters for each agent and each objective can avoid coupling but may fail to control system-level worst-case risk because “bad events” can move between channels over time.

This paper studies distribution-free risk control in the presence of *multiple channels*—losses indexed by agent and objective—and focuses on a worst-channel metric that captures the realized frequency and severity of the most critical safety events in the system. We propose a controller that retains a clean worst-channel guarantee but localizes conservatism in space and time, avoiding global slowdown.

a) Contributions.:

- We formalize distribution-free worst-channel risk control for online multi-agent decision systems with multiple safety objectives.
- We prove by construction that independent per-channel calibration does not control the time-average of the per-time maximum loss.
- We propose a *selective system-level controller*: a single parameter is updated using the realized worst-channel loss, but it is applied only to channels that were recently maximal; other channels may retain local parameters for efficiency.
- Under a mild delayed-safety condition, we derive a finite-time bound on the worst-channel empirical risk. We instantiate the framework in a multi-drone navigation simulation with inertial dynamics, short-horizon prediction, and spline planning, specifying how the abstract theory maps to this applied setting.

II. RELATED WORK

Our work is inspired by recent decision-focused frameworks that provide distribution-free safety guarantees by calibrating a decision parameter online. The closest reference is Conformal Decision Theory [1], which proposes feedback controllers that control empirical risk for a *single* loss channel under adversarial sequences. We adopt the same adversarial-sequence mindset and emphasize that the resulting guarantees do not rely on exchangeability or a probabilistic model. Our contribution is a multi-agent, multi-objective extension that targets a *worst-channel* metric while avoiding global coupling.

The adversarial-sequence model is standard in online learning and online convex optimization [2], [3]. In that literature, guarantees are often expressed as regret relative

to a comparator. Here, we focus on direct control of a safety metric on the realized sequence, which is more aligned with safety requirements that are not naturally posed as regret minimization. In robotics, safety is commonly formalized via robust constraints, chance constraints, or control barrier functions; these approaches typically require distributional assumptions, conservative uncertainty sets, or known dynamics. The distribution-free perspective complements these tools: it provides a transparent retrospective guarantee that can sit on top of an existing planner, regardless of its internal structure.

III. PROBLEM FORMULATION

We consider an online decision-making system over discrete times $t = 1, 2, \dots, T$. At time t , the system observes an input $x_t \in \mathcal{X}$, selects an action $u_t \in \mathcal{U}$, and an outcome $y_t \in \mathcal{Y}$ is realized. The system incurs a bounded loss $L(u_t, y_t) \in [0, 1]$. We make no stochastic assumptions on (x_t, y_t) : the sequence may be nonstationary, non-exchangeable, and even adversarial [2], [3].

The system does not select arbitrary actions; rather, at each time t it has access to a family of decision rules indexed by a scalar parameter:

$$\mathcal{D}_t := \{D_t^\lambda : \mathcal{X} \rightarrow \mathcal{U} \mid \lambda \in \mathbb{R}\}. \quad (1)$$

The deployed action is $u_t = D_t^{\lambda_t}(x_t)$ where λ_t is chosen online. Our objective is to control the realized empirical risk

$$\hat{R}_T(\lambda_{1:T}) := \frac{1}{T} \sum_{t=1}^T L(D_t^{\lambda_t}(x_t), y_t) \quad (2)$$

at or below a target level $\varepsilon \in [0, 1]$ up to terms that vanish with T .

A. Multi-agent, multi-objective channels

In multi-agent settings with multiple objectives, it is natural to define a set of *channels* $\mathcal{C}$ indexing distinct safety losses. At time t , each channel $c \in \mathcal{C}$ incurs a bounded loss $\ell_t^c \in [0, 1]$. The system-level *worst-channel* loss is

$$m_t := \max_{c \in \mathcal{C}} \ell_t^c \in [0, 1], \quad (3)$$

and the retrospective metric of interest is

$$\hat{R}_T^{\max} := \frac{1}{T} \sum_{t=1}^T m_t. \quad (4)$$

This “average of the per-time maximum” captures the realized frequency and severity of the most critical safety events across the system.

We emphasize that $\mathcal{C}$ may include channel losses that are not independent: for example, a pairwise separation channel depends on the joint actions of two agents. Our analysis places no restrictions on these dependencies; they are absorbed into the decision rules D_t^λ and the realized losses.

IV. WHY INDEPENDENT PER-CHANNEL CALIBRATION FAILS

A common baseline is independent calibration: each channel c maintains its own parameter and aims to control its own average loss $\frac{1}{T} \sum_t \ell_t^c$. This avoids coupling, but it does not control the worst-channel objective (4).

Proposition 1 (Independent control does not imply worst-channel control). *For any $\varepsilon \in (0, 1)$ and any integer $N \geq 2$, there exist bounded loss sequences $\ell_t^1, \dots, \ell_t^N \in [0, 1]$ such that each channel satisfies $\frac{1}{T} \sum_{t=1}^T \ell_t^i \leq \varepsilon$ while the worst-channel metric satisfies $\hat{R}_T^{\max} = 1$.*

Proof. Partition the time indices $\{1, \dots, T\}$ into N disjoint blocks of length εT (assume εT is an integer and $N\varepsilon \leq 1$; otherwise truncate the final block). In block i , set $\ell_t^i = 1$ and set $\ell_t^k = 0$ for all $k \neq i$. Each channel is active for exactly εT steps, hence $\frac{1}{T} \sum_{t=1}^T \ell_t^i = \varepsilon$ for all i . However, at every time step exactly one channel has loss one, so $m_t = \max_i \ell_t^i = 1$ for all t and therefore $\hat{R}_T^{\max} = \frac{1}{T} \sum_{t=1}^T 1 = 1$. $\square$

Proposition 1 captures the key failure mode: worst events can move between channels over time. Controlling each channel's average loss does not prevent the system from suffering a near-violation almost every step, because the identity of the maximal channel may change arbitrarily. This motivates controllers that explicitly track and respond to the system-level worst-channel loss.

V. SELECTIVE SYSTEM-LEVEL RISK CONTROL

We now describe a controller that targets $\hat{R}_T^{\max}$ while avoiding global slowdown. The core idea is to maintain a *single* system-level parameter λ_t^G that is updated using the realized worst-channel loss m_t , but to apply this parameter only to a subset of channels that are likely to dominate worst-channel risk. Channels outside this subset can retain local parameters for efficiency.

A. Active sets and effective parameters

Let $A_t \subseteq \mathcal{C}$ be an *active set* chosen using only past information (measurable with respect to the history up to time $t - 1$). A simple and effective choice is to use the channels that were maximal at the previous step:

$$A_t \subseteq \arg \max_{c \in \mathcal{C}} \ell_{t-1}^c. \quad (5)$$

More generally, A_t may contain the top- k channels, channels exceeding a threshold, or channels predicted to become critical by a model; the analysis below does not depend on the specific choice, only on measurability with respect to past information.

Each channel c may maintain a local parameter λ_t^c . Define the *effective* parameter used at time t :

$$\tilde{\lambda}_t^c := \begin{cases} \lambda_t^G, & c \in A_t, \\ \lambda_t^c, & c \notin A_t. \end{cases} \quad (6)$$

The decision rule $D_t^{\lambda_t^G}$ in (1) is interpreted as “run the (possibly coupled) planner using $\tilde{\lambda}_t^c$ for each channel c .” In other words, λ_t^G affects planning only through the channels in A_t .

Algorithm 1 Selective system-level risk control

- 1: Initialize λ_1^G , local $\{\lambda_1^c\}_{c \in \mathcal{C}}$, step sizes η, η_{loc} , target ε .
 - 2: **for** $t = 1$ to T **do**
 - 3: Choose active set A_t from past losses (e.g., (5)).
 - 4: Set effective parameters $\tilde{\lambda}_t^c$ via (6).
 - 5: Execute $u_t = D_t^{\lambda_t^G}(x_t)$ (planner uses $\tilde{\lambda}_t^c$ internally).
 - 6: Observe channel losses $\{\ell_t^c\}_{c \in \mathcal{C}}$ and set $m_t = \max_c \ell_t^c$.
 - 7: Update $\lambda_{t+1}^G \leftarrow \lambda_t^G + \eta(\varepsilon - m_t)$.
 - 8: Optionally update locals via (8).
 - 9: **end for**
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B. Update rule

The system-level parameter is updated by one-dimensional feedback:

$$\lambda_{t+1}^G = \lambda_t^G + \eta(\varepsilon - m_t), \quad (7)$$

where $\eta > 0$ is a step size. This controller structure has been shown to control empirical risk in a fully distribution-free sense for a single channel [1]; here we apply it to the aggregated loss m_t .

Local parameters may be held fixed, learned by other mechanisms, or updated independently. In the drone application we use a clipped local feedback update for stability:

$$\lambda_{t+1}^c \leftarrow \Pi_{[\lambda_{\min}, \lambda_{\max}]}(\lambda_t^c + \eta_{\text{loc}}(\varepsilon - \ell_t^c)), \quad (8)$$

with step size $\eta_{\text{loc}} > 0$ and projection Π . Local updates improve efficiency but are not needed for the worst-channel guarantee.

C. Design remarks

Two points are worth emphasizing.

Global guarantee, local effect. The update (7) controls a system-level metric, so its bound is not diluted by the number of channels. At the same time, selective application through (6) ensures that the *behavioral* effect of the global parameter is localized to channels that appear to drive worst-channel risk.

Active-set measurability. The analysis requires only that A_t be chosen from past information (i.e., before observing y_t). This allows a wide range of policies: previous maximizers, top- k channels, thresholding, or learned predictors that anticipate which channels will become critical. The guarantee does not change with the policy; only the induced losses m_t change.

Projection in practice. In implementation it is common to project λ_t^G to a bounded interval $[\lambda_{\min}, \lambda_{\max}]$ to avoid numerical blow-up and to cap conservatism. The theoretical analysis remains unchanged as long as the “safe” region (the threshold λ_{safe} in Definition 1) is contained in the projection interval. We use projection in the simulation, but omit it from the core update for notational clarity.

Step-size tradeoff. Larger η reacts faster to large losses but increases steady-state oscillations; smaller η yields smoother behavior but responds more slowly. Theorem 1 makes this explicit: the transient term scales as $(\lambda_1^G - \lambda_{\text{safe}})/(\eta t)$ while the delay term scales as K/t .

VI. A FINITE-TIME WORST-CHANNEL GUARANTEE

We now derive a distribution-free guarantee for the worst-channel metric (4). The analysis treats the entire multi-agent system as a single online decision-maker with scalar control variable λ_t^G ; the complexity of the planner, prediction model, coupling between agents, and the active-set mechanism is absorbed into the time-varying decision families $(\mathcal{D}_t)$.

A. Delayed safety

The key assumption is that sufficiently conservative parameters eventually lead to low average worst-channel loss, possibly after a transient period.

Definition 1 (Delayed safety). *The decision families $(\mathcal{D}_t)_{t=1}^T$ are delayed-safe for the worst-channel loss if there exist an integer $K \geq 1$, a threshold $\lambda_{\text{safe}} \in \mathbb{R}$, and a level $\varepsilon_{\text{safe}} \leq \varepsilon$ such that for all t_0 ,*

$$\left(\forall t \in \{t_0, \dots, t_0 + K - 1\}, \lambda_t^G \leq \lambda_{\text{safe}} \right) \Rightarrow \frac{1}{K} \sum_{t=t_0}^{t_0+K-1} m_t \leq \varepsilon_{\text{safe}} \quad (9)$$

Delayed safety is distribution-free and makes no probabilistic claims. It is a property of the decision families, the loss definition, and the environment. When $K = 1$ it reduces to instantaneous safety; allowing $K > 1$ is crucial in dynamic systems with inertia.

B. Lower bound on the control variable

Proposition 2 (Uniform lower bound). *Assume $m_t \in [0, 1]$ and run the update (7) with $\varepsilon \in [0, 1]$. If $\lambda_1^G \geq \lambda_{\text{safe}} - \eta$ and the system is delayed-safe (Definition 1), then*

$$\lambda_t^G \geq \lambda_{\text{safe}} - K\eta \quad \text{for all } t \in \{1, \dots, T + 1\}. \quad (10)$$

Proof. Because $m_t \in [0, 1]$ and $\varepsilon \in [0, 1]$, each update changes λ^G by at most η : $|\lambda_{t+1}^G - \lambda_t^G| \leq \eta$. Suppose for contradiction that $\min_{s \leq T+1} \lambda_s^G < \lambda_{\text{safe}} - K\eta$. Let

$$t := \arg \min \{s \in \{1, \dots, T + 1\} : \lambda_s^G < \lambda_{\text{safe}} - K\eta\}$$

be the first time the parameter drops below $\lambda_{\text{safe}} - K\eta$. By the per-step change bound, for each $k \in \{1, \dots, K\}$ we have $\lambda_{t-k}^G < \lambda_{\text{safe}} - (K - k)\eta \leq \lambda_{\text{safe}}$. Thus, the K -step window $\{t - K, \dots, t - 1\}$ lies entirely in the safe region, and delayed safety (9) implies $\frac{1}{K} \sum_{s=t-K}^{t-1} m_s \leq \varepsilon_{\text{safe}} \leq \varepsilon$. Unrolling the update over these K steps gives

$$\lambda_t^G = \lambda_{t-K}^G + K\eta \left(\varepsilon - \frac{1}{K} \sum_{s=t-K}^{t-1} m_s \right) \geq \lambda_{t-K}^G.$$

But t is the first time the sequence falls below $\lambda_{\text{safe}} - K\eta$, so $\lambda_{t-K}^G \geq \lambda_{\text{safe}} - K\eta$, contradicting $\lambda_t^G < \lambda_{\text{safe}} - K\eta$. $\square$

C. Main theorem

Theorem 1 (Finite-time worst-channel bound). *Assume $m_t \in [0, 1]$ for all t . Run Algorithm 1 with step size $\eta > 0$ and target $\varepsilon \in [0, 1]$. Suppose delayed safety (Definition 1) holds with parameters $(K, \lambda_{\text{safe}}, \varepsilon_{\text{safe}} \leq \varepsilon)$, and that $\lambda_1^G \geq \lambda_{\text{safe}} - \eta$. Then for all $t \geq K$,*

$$\hat{R}_t^{\max} = \frac{1}{t} \sum_{s=1}^t m_s \leq \varepsilon + \frac{\lambda_1^G - \lambda_{\text{safe}}}{\eta t} + \frac{K}{t}. \quad (11)$$

Proof. Unrolling (7) gives

$$\lambda_{t+1}^G = \lambda_1^G + \eta \sum_{s=1}^t (\varepsilon - m_s).$$

Rearranging yields the identity

$$\frac{1}{t} \sum_{s=1}^t m_s = \varepsilon + \frac{\lambda_1^G - \lambda_{t+1}^G}{\eta t}.$$

Proposition 2 gives $\lambda_{t+1}^G \geq \lambda_{\text{safe}} - K\eta$, which implies (11). $\square$

Corollary 1 (Asymptotic control). *Under the assumptions of Theorem 1, we have*

$$\limsup_{t \rightarrow \infty} \hat{R}_t^{\max} \leq \varepsilon.$$

a) *Interpretation.*: Theorem 1 is fully distribution-free and planner-agnostic. The active-set mechanism is absorbed into the decision family: for each fixed λ , the rule D_t^λ can incorporate arbitrary logic, including the active set, prediction, and local parameters. The only substantive modeling requirement is delayed safety, which is natural when dynamics prevent instantaneous avoidance.

VII. APPLICATION: MULTI-DRONE NAVIGATION AMONG PEDESTRIANS

We now instantiate the abstract framework in a concrete and visually interpretable scenario: a team of drones must repeatedly traverse a crowded area populated by moving pedestrians while maintaining (i) human separation and (ii) inter-drone separation. The application is designed to make three challenges explicit: (a) prediction is imperfect but decision-relevant; (b) dynamics induce delay so conservatism cannot take effect instantly; and (c) worst-channel events are sparse, moving between channels over time.

A. Environment and scenario design

We simulate a planar arena of size 55 m $\times$ 55 m with $N = 10$ drones and $M = 46$ pedestrians. Pedestrians follow smooth trajectories with short-term approximately constant velocities plus small stochastic heading perturbations, producing stable flows and counterflows while keeping short-horizon prediction plausible. To increase decision-relevant interaction without resorting to chaotic motion, a fraction of pedestrians drift slowly in a central “plaza” region, creating localized congestion and intermittent near-misses. Each drone is repeatedly assigned a goal on the opposite side of the arena. Upon reaching its goal, a new goal is sampled on the far side, creating sustained traffic and repeated interactions over long episodes.

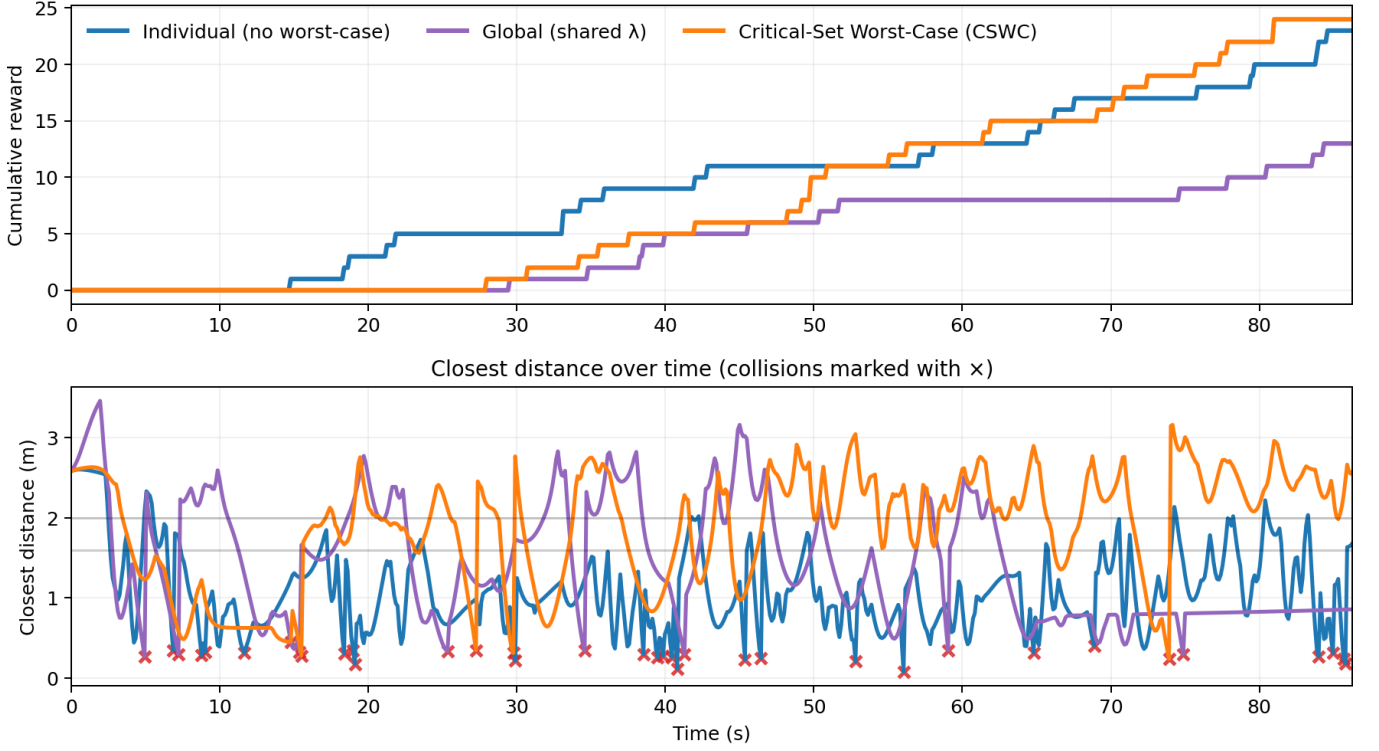


Fig. 2: Application performance. *Top:* cumulative reward (successful goal deliveries) over time. *Bottom:* closest separation distance between any drone–pedestrian or drone–drone pair. Red $\times$ markers indicate collision events (drone “deaths”), computed at the collision timestep before respawn-induced distance jumps. GLOBAL uses a shared λ and becomes conservative whenever any agent encounters risk; INDIVIDUAL adapts per-agent λ values but fails to calibrate worst-case risk and collides frequently; CSWC applies the worst-case λ to the worst channel and to predicted critical-set incursions, yielding high reward with few or no collisions.

B. Drone dynamics (inertia)

Each drone has position $p_t \in \mathbb{R}^2$ and velocity $v_t \in \mathbb{R}^2$. The planner chooses a desired velocity v_t^{des} , which the drone tracks with a first-order response:

$$\begin{aligned} a_t &= \text{clip}\left(\frac{v_t^{\text{des}} - v_t}{\tau}, a_{\max}\right), \\ v_{t+1} &= \text{clip}(v_t + \Delta t a_t, v_{\max}), \\ p_{t+1} &= p_t + \Delta t v_{t+1}. \end{aligned} \quad (12)$$

Here τ is a time constant, $a_{\max}$ and $v_{\max}$ are acceleration and speed limits, and $\text{clip}(z, r)$ clips the vector norm of z to r . This produces momentum-like behavior: drones cannot instantaneously stop or turn, which is essential both for realism and for motivating delayed safety with $K > 1$.

C. Prediction model and predictive critical sets

At each step the planner predicts pedestrian motion over a short horizon of $H = 20$ steps (approximately 2.4 seconds) using constant-velocity extrapolation. This model is intentionally imperfect: when pedestrians change direction, prediction error can create a mismatch between planned and realized distances. Following the discussion in Section VII, we use *predictive critical sets* to trigger conservatism early enough to

compensate for inertia: a drone is treated as “critical” for human separation if its *predicted* minimum distance to any pedestrian within the next H steps falls below the 2 m separation threshold. This preserves the physical meaning of the safety margin while allowing the controller to act before the violation becomes unavoidable.

D. Safety channels and bounded losses

We define two safety objectives:

- **Human separation:** each drone should maintain distance at least $d_{\text{ped}} = 2$ m from all pedestrians.
- **Inter-drone separation:** each pair of drones should maintain distance at least $d_{\text{dd}} = 1.6$ m.

For drone i at time t , let $d_{t,\min}^{(i,\text{ped})}$ be the minimum distance to any pedestrian and let $d_{t,\min}^{(i,\text{dd})}$ be the minimum distance to any other drone. We use normalized, bounded margin violations:

$$\begin{aligned} \ell_t^{(i,\text{ped})} &= \min\left(1, \max\left(0, \frac{d_{\text{ped}} - d_{t,\min}^{(i,\text{ped})}}{d_{\text{ped}}}\right)\right), \\ \ell_t^{(i,\text{dd})} &= \min\left(1, \max\left(0, \frac{d_{\text{dd}} - d_{t,\min}^{(i,\text{dd})}}{d_{\text{dd}}}\right)\right). \end{aligned} \quad (13)$$

Channels are indexed as $c = (i, \text{ped})$ and $c = (i, \text{dd})$ for $i \in [N]$. The worst-channel loss is $m_t = \max_i \max(\ell_t^{(i, \text{ped})}, \ell_t^{(i, \text{dd})})$.

E. Decision family: short-horizon velocity planning

Each drone chooses a desired velocity by minimizing a short-horizon objective over a discrete candidate set of headings and speeds. For each candidate v^{des} , we roll out the inertial dynamics (12) for H steps under predicted pedestrian and drone motion and compute:

$$J = J_{\text{goal}} + w_{\text{ped}} J_{\text{ped}} + w_{\text{dd}} J_{\text{dd}},$$

where J_{goal} encourages progress to the current goal and $J_{\text{ped}}, J_{\text{dd}}$ aggregate predicted proximity violations along the rollout (e.g., sums of squared hinge losses). Penalty weights are monotone functions of *effective* parameters (Section VII), with smaller λ corresponding to larger penalties (more caution). For visualization, the planned horizon trajectory is rendered as a spline through the rolled-out positions.

F. Active sets and “worst-channel always” application

We compare three controllers whose *planner and dynamics are identical*; they differ only in how conservatism parameters are adapted and applied:

- 1) **Global shared parameter:** one scalar λ_t^G is updated using m_t and applied to *all* channels at all times. This yields a worst-channel guarantee but couples the entire system: one near-miss forces everyone to become cautious.
- 2) **Independent calibration:** per-channel parameters are updated by their own losses. This avoids coupling but, by Proposition 1, does not control the worst-channel metric.
- 3) **Selective worst-channel control (CSWC):** per-channel parameters are maintained, and a fixed “worst-case” parameter λ_G is applied (i) to channels in the predictive critical set and (ii) *always* to the worst instantaneous channel (including both drones in the closest inter-drone pair), even if it is not currently critical. This implements the principle “global update, local effect” while explicitly tracking the channel that dominates m_t .

G. Results: efficiency–safety tradeoff in long-horizon interaction

Table I reports a representative long episode (seed fixed for reproducibility) using the metrics that align with our theory and with human interpretability: cumulative reward (goal completions) and collision count (hard contacts leading to delayed respawn). The qualitative behavior matches the intended story. The global controller suppresses worst events by coupling everyone to the same conservative parameter, which reduces throughput. Independent calibration attains high throughput but suffers frequent collisions as “bad events” move across channels. CSWC achieves the best of both worlds: it maintains high throughput while reducing collisions by localizing conservatism to predicted-critical and worst-case channels.

TABLE I: Drone–pedestrian simulation outcomes (arena 55×55 m, $N = 10$ drones, $M = 46$ pedestrians, horizon $H = 20$ steps ≈ 2.4 s). “Reward” counts goal completions; “Collisions” counts hard contacts leading to delayed respawn.

Controller	Reward $\uparrow$	Collisions $\downarrow$
Independent calibration	23	23
Global shared parameter	13	8
Selective worst-channel (CSWC)	24	5

H. Why delayed safety is plausible here

Delayed safety is a statement about the existence of a sufficiently conservative regime. In this simulation, when the effective parameter for an active channel becomes small enough, the corresponding penalty weights grow and the planner favors actions that increase separation margins. Because the drones have inertia, it may take multiple steps for these actions to produce realized separation. This is precisely the behavior captured by delayed safety with $K > 1$: conservatism needs time to take effect, and predictive critical sets help ensure the controller “gets ahead” of the dynamics rather than reacting after the fact.

VIII. EXPERIMENTAL PROTOCOL AND BASELINES

We recommend comparing four controllers:

- 1) **No risk control:** fixed penalty weights (equivalently, fixed λ).
- 2) **Global shared parameter:** one λ_t^G applied to all channels at all times, updated by m_t .
- 3) **Independent calibration:** per-channel parameters updated by their own losses, with no system-level update.
- 4) **Selective system-level control (ours):** Algorithm 1.

a) *Metrics.*: We report the realized worst-channel risk $\hat{R}_T^{\text{max}}$, collision and near-collision counts, and efficiency measures such as time-to-goal and path length. Since the guarantee concerns $\hat{R}_T^{\text{max}}$, the key quantitative question is whether selective control reduces the frequency of the largest safety violations without imposing a uniform slowdown.

b) *Qualitative presentation.*: Qualitatively, the global shared parameter tends to reduce worst events by slowing everyone, while selective control reduces worst events by tightening only the channels that repeatedly dominate worst-case risk, preserving efficiency elsewhere. In a website demonstration, the most persuasive story is visual: for long stretches the baseline and selective controller behave similarly, then at the moments where prediction error and inertia create a near-violation, the selective controller temporarily increases conservatism only for the implicated drone(s) and objective(s). Rendering short “ghost” rollouts of the no-risk-control trajectory for a few steps after each evasive maneuver highlights precisely when (and how) risk control intervenes.

c) *Hard collisions.*: For visualization realism, drones are treated as hard disks. If a pairwise distance falls below a small radius, we resolve overlap by projecting positions along the separating axis and damping relative velocity so that agents cannot pass through each other. This implementation

detail affects only the realized losses and trajectories; the distribution-free guarantee continues to apply because it makes no assumptions beyond bounded losses and delayed safety.

IX. CONCLUSION

We presented a distribution-free approach to risk control for online multi-agent decision systems with multiple safety objectives. Our focus was the worst-channel retrospective metric, which captures the realized frequency and severity of the most critical safety events across the system. We showed that independent per-channel calibration cannot control this metric in general, because worst events can move between channels over time.

To address this, we proposed a selective system-level controller: a single scalar parameter is updated using the realized worst-channel loss, but it is applied only to channels that were recently maximal. This design preserves a clean, distribution-free finite-time guarantee under a mild delayed-safety condition, while avoiding the practical pathology of globally slowing all agents in response to localized rare events. We then mapped the abstract framework to a multi-drone navigation simulation with inertial dynamics, short-horizon prediction, and spline visualization, illustrating how the theoretical objects (channels, decision families, delayed safety) appear in a concrete deployment pipeline.

Several directions are immediate. First, richer active-set policies that use learned predictors to anticipate which channels will become critical could further improve efficiency while leaving the worst-channel guarantee unchanged, since it depends only on the realized worst-channel loss. Second, extending the analysis to multiple simultaneous system-level metrics (for example, separate worst-channel targets for distinct objective groups) would better match deployments where some objectives are strictly safety-critical while others are soft. Finally, evaluating the controller on physical robotic platforms would clarify how delayed safety manifests under real sensing and actuation constraints and would help connect the abstract parameter dynamics to practical tuning guidelines.

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